

Epigenetic Adaptation Drives Monocyte Differentiation into Microglia-Like Cells Upon Engraftment into the Central Nervous System

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eLife Assessment

The authors investigated the epigenetic mechanisms regulating the differentiation of circulating monocytes that infiltrate the CNS and adopt microglia-like characteristics. The work is **useful** to the field, as the contribution of circulating myeloid cell-derived microglia remains controversial. However, the evidence presented is **inadequate** as the analyses are based on a very limited set of genes, which does not sufficiently support the authors' central claims.

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Abstract

The identification of specific markers to distinguish resident microglia from infiltrating monocytes has been a long-standing challenge in neuroscience. Recently, proteins such as P2RY12, TMEM119, and FCRL5 have been proposed as microglia-specific and are now widely used to define microglial populations in health and disease. The specificity of these markers was predicated on the assumption that circulating monocytes retain their distinct signatures after entering the central nervous system (CNS). Here, we challenge this paradigm. Using a combination of bone marrow chimeras, single-cell RNA sequencing, ATAC-seq, flow cytometry, and immunohistochemistry, we demonstrate that monocytes engrafting into the CNS acquire de novo expression of these established microglia markers. This phenotypic conversion is driven by profound epigenetic reprogramming, characterized by dynamic changes in chromatin accessibility at key gene loci, including P2ry12, Tmem119, and Aif1 (Iba1), and a shift in transcription factor binding motifs toward a microglial profile. We show

this process occurs in the retina following injury and, remarkably, under physiological conditions in the brain and spinal cord, where blood-derived monocytes progressively contribute to the resident myeloid pool. Furthermore, engrafted monocytes downregulate canonical monocyte markers (Ly6C, CD45), eventually becoming indistinguishable from embryonic microglia based on conventional phenotyping. Our findings reveal that infiltrating monocytes undergo extensive epigenetic and transcriptional remodeling to adopt a microglia-like fate, challenging the specificity of current markers and necessitating a re-evaluation of the distinct roles of these two cell populations in CNS pathology.

Significance Statement

Distinguishing resident CNS microglia from infiltrating monocytes is fundamental to understanding neuro-inflammation. This study reveals that widely used “microglia-specific” markers are not exclusive, as monocytes entering the central nervous system are epigenetically reprogrammed to express them. This mimicry invalidates long-held assumptions about microglial identity and demonstrates that many cells previously identified as microglia may have a peripheral origin. Our work underscores the critical need for more reliable methods to differentiate these populations to accurately define their respective contributions to CNS health and disease.

Introduction

Microglia and infiltrating peripheral monocytes are central players in the pathology of central nervous system (CNS) diseases and have emerged as key therapeutic targets [1–5]. However, given their overlapping functions and morphology, distinguishing these two myeloid populations is notoriously difficult. The use of inadequately specific markers often leads to inadvertent mislabeling, compromising the interpretation of experimental data and our understanding of neuro-inflammatory processes.

Recently, a new panel of markers, including P2RY12, TMEM119, and FCRL5, was proposed to be exclusively expressed by microglia and has been widely adopted [6, 7]. The presumed specificity of these markers was based on comparing CNS-resident microglia with non-CNS monocytes or purified acute immune cells of the CNS [6, 7]. This framework, however, largely overlooked the possibility that monocytes might alter their identity after permanently integrating into the CNS parenchyma. Indeed, recent findings indicate that infiltrating monocytes undergo persistent phenotypic changes upon engraftment, adopting a ramified, microglia-like morphology while retaining a distinct, pro-inflammatory capacity that can drive disease [2, 8].

Here, we hypothesized that peripheral monocytes undergo profound epigenetic reprogramming upon CNS engraftment, enabling them to express canonical microglia markers and confounding their identification. We investigated this hypothesis in the retina—a well-established model of the CNS—and extended our findings to the brain and spinal cord. We demonstrate that shortly after engraftment, monocytes initiate expression of P2RY12, TMEM119, and FCRL5. This phenotypic switch is underpinned by dynamic changes in chromatin accessibility and the acquisition of a microglia-associated transcriptional network, including key transcription factors like PU.1, IRF2, and CEBP. Concurrently, these cells suppress canonical monocyte markers (CCR2, Ly6C, CD45), eventually becoming phenotypically indistinguishable from embryonic microglia by conventional methods.

Our study highlights the remarkable plasticity of monocytes and reveals a fundamental flaw in the tools currently used to identify them. We further demonstrate that the rules governing monocyte engraftment differ across CNS compartments; while injury is required for retinal infiltration, monocytes physiologically populate the brain and spinal cord over time. These findings raise a critical scientific question: how can we reliably discriminate between embryonic microglia and these newly identified monocyte-derived “microglia-like” cells? Answering this is essential to finally untangle the distinct roles of these two crucial immune populations in CNS pathology.

Results

Single-cell RNAseq reveals a transcriptional shift of engrafted monocytes towards a microglia signature

To track the transcriptional fate of myeloid cells after injury, we performed single-cell RNA-seq on isolated CD45⁺CD11b⁺ cells from mouse retinas at baseline and at 1, 4, and 7 days after a corneal alkali injury, a model known to induce monocyte infiltration and engraftment [2, 9, 10]. We identified four distinct cell clusters (Fig. 1A). The baseline cluster (Cluster 1) corresponds to native, yolk-sac-derived microglia, as the healthy retina is not populated by peripheral monocytes [2, 8]. By day 1 post-injury, two separate clusters emerged: a monocyte cluster (Cluster 3), identified by high expression of the monocyte marker *Siglec1*, and a microglia cluster (Cluster 4) with low *Siglec1* expression (Fig. 1B) [11]. Remarkably, by day 4, these populations merged into a single transcriptional cluster (Cluster 2), and by day 7, all cells resolved into the original naive microglia cluster (Cluster 1), demonstrating a dynamic transcriptional convergence towards a microglia signature (Fig. 1A).

Gene expression analysis showed that the classical macrophage marker *Cx3cr1* was uniformly expressed, while the monocyte marker *Ccr2* was predominantly expressed in the monocyte-rich cluster 3 (day 1) and the mixed cluster 2 (day 4) (Fig. 1C, D). Critically, the putative microglia markers *p2ry12*, *tmem119*, and *fcrls* were acquired by the engrafting population over time. While *tmem119* and *fcrls* were expressed broadly, *p2ry12* expression was absent in early infiltrating monocytes (day 1) but became highly expressed by day 4 and 7, mirroring the pattern of naive microglia (Fig. 1C, D). *Aif1* (or *iba1*) was expressed in all clusters and highly in cluster 2 (day 4). *P2ry12* expression exhibited temporal regulation in monocytes, with lack of expression in clusters 3 and 4 – both representing early-stage infiltration (Day 1), confirmed in protein studies (Fig. 1C, D). Using a CX3CR1^{+/EGFP}::CCR2^{+/RFP} bone marrow chimera (BMT) model [2, 4, 12], we confirmed via flow cytometry that infiltrating monocytes gradually downregulate their CD45 expression, and by day 7, their CD45 profile becomes indistinguishable from that of resident microglia, further corroborating the scRNAseq findings of a converging cellular signature (Fig. 1E). The BMT model had high transplantation efficiency with 93.6% of the blood CD45⁺CD11b⁺CX3CR1⁺ cells from recipient mice being EGFP⁺RFP⁺ (Fig. S1). The expected profile of CD45 expression of embryonic microglia was assessed using a transgenic CX3CR1^{+/EGFP} mouse without injury.

By day seven post-injury, flow cytometry confirmed that engrafted CX3CR1⁺ monocytes modulated their CD45 expression to levels indistinguishable from resident microglia, corroborating the scRNA-seq data indicating a converging cellular signature (Fig. 1E). We further validated these transcriptional changes using qPCR on sorted cells from bone marrow chimeras 45 days after engraftment. This analysis confirmed that engrafted monocytes acquire mRNA for *p2ry12*, *tmem119*, *fcrls*, and *aif1* (*Iba1*). Notably, while *p2ry12* mRNA in engrafted monocytes was significantly upregulated compared to blood monocytes, its expression remained lower than that of either naïve or injured microglia. In contrast, resident microglia maintained consistently high expression of these markers in both naïve and injured conditions (Fig. S2).

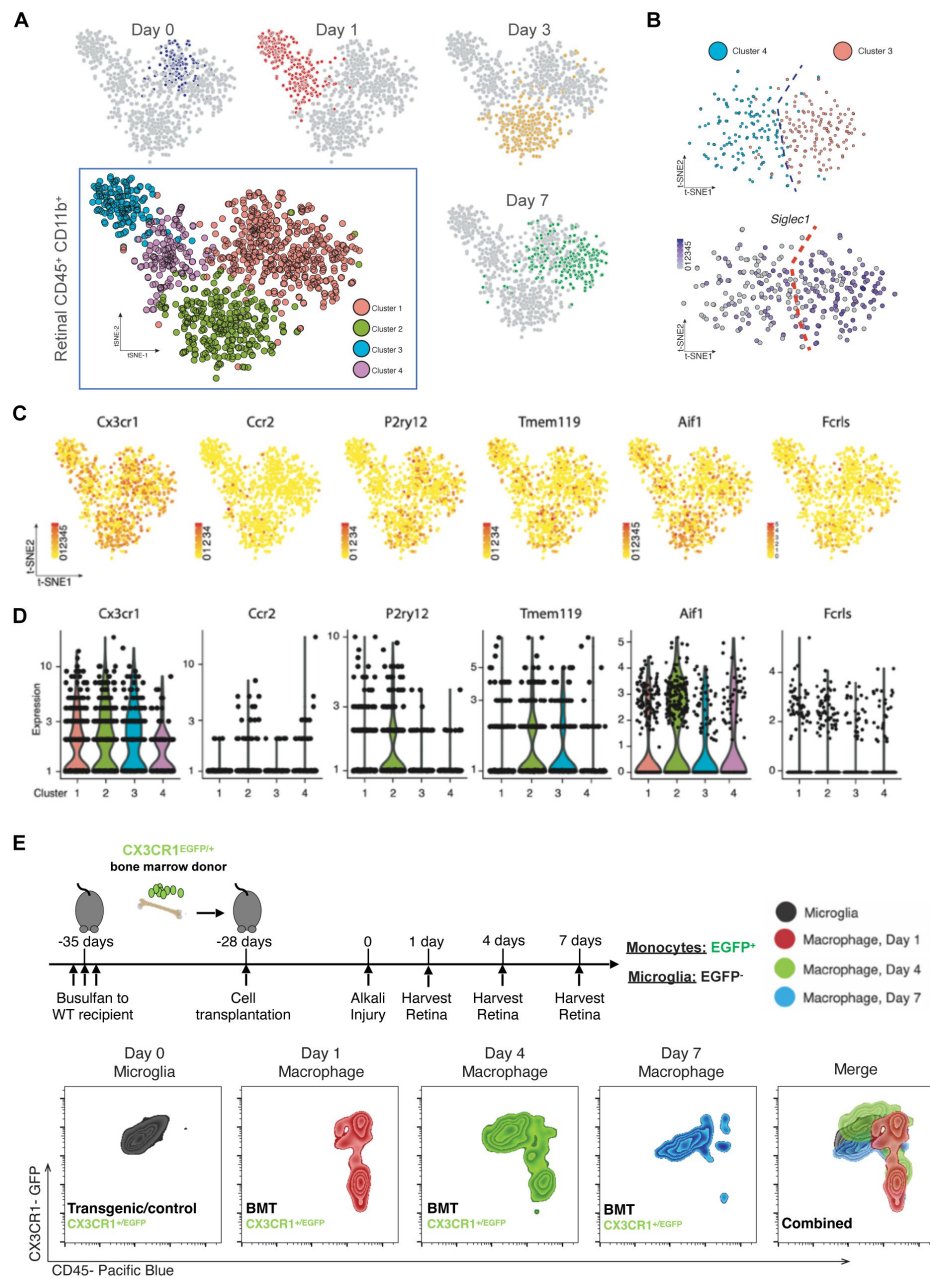


Figure 1.

Single-cell RNAseq analysis for retinal CD45⁺CD11b⁺ cells 0, 1, 4 and 7 days after ocular injury.

(A) Principal component analysis shows the existence of 4 clusters with distinct transcriptional profiles assigned to the day of cell retrieval. CD45⁺CD11b⁺ cells undergo extensive changes in their transcriptome during retinal engraftment and by day 7 acquire an identical signature to naïve microglia. **(B)** Siglec1⁺ gene is expressed in clusters 3 and 4, both assigned to day 1, suggestive of the presence of monocytes within the microglia sample. **(C)** tSNE analysis of *cx3cr1*, *ccr2*, *p2ry12*, *tmem119*, *aif1*, and *fcrls* expression in cluster 4 and **(D)** graphical representation of the expression of microglia markers *p2ry12*, *tmem119*, *AIF1*, and *fcrls* in the 4 clusters, including those representatives of microglia and monocytes signatures. **(E)** Flow cytometric analysis of the expression of CD45 marker in retinal CX3CR1⁺ cells before and 1, 4, and 7 days after injury. A transgenic CX3CR1⁺/EGFP mouse was used to show the expected CD45 profile of pure microglia from a non-chimera and without injury, while the BMT panels show the evolution in the experimental model. As reference we used a CX3CR1⁺/EGFP mouse, stained with CD45 markers. Double positive CD45⁺ CX3CR1⁺ cells represent only microglia, since naïve eyes do not have infiltration of monocytes [24]. To map infiltrating/engrafting monocytes, we used a CX3CR1⁺/EGFP bone marrow chimera. CX3CR1⁺ CD45⁺ infiltrating monocytes gradually transitioned their CD45 expression towards the expression of naïve microglia at 7 days.

Engrafted monocytes acquire de novo protein expression of P2RY12, FCRLS, and TMEM119

We next sought to confirm if these transcriptional changes translate to the protein level. Using CX3CR1^{+/EGFP}::CCR2^{+/RFP} bone marrow chimeras to distinguish donor monocytes (GFP⁺) from resident microglia (GFP⁻), we analyzed protein expression in the retina at multiple time points after injury (**Fig. 2A**). We first verified that circulating blood monocytes lack P2RY12, TMEM119, and IBA1 protein expression (**Fig. S3**). We also established the protein expression baseline for circulating monocytes prior to CNS engraftment. Flow cytometry analysis of CD45⁺CD11b⁺CX3CR1⁺ blood cells from naïve mice confirmed the absence of FCRLS protein (**Fig. 2D**) and the robust MHC-II protein expression, which served as a positive staining control (**Fig. S3**).

Following retinal infiltration, engrafted monocytes (EGFP⁺) began to express these markers. P2RY12 protein was absent at days 1 and 7 (white arrow) but became detectable by day 14 (**Fig. S4**) and was expressed in approximately 55% of engrafted monocytes by day 45, which by then had adopted a ramified, microglia-like morphology (**Fig. 2B**, **Fig. S4**). This heterogeneous expression was sustained long-term, at 16 months post-engraftment, as assessed in a CX3CR1^{CreER}::ROSA26^{tdTomato} bone marrow chimera model (**Fig. S5 A, B**), while being consistently below the expression from embryonic microglia. In contrast, TMEM119 expression was rapidly acquired, with 25% of monocytes positive at day 1 and 100% positive by day 7, an expression level that was maintained at 45 days and 16 months (**Fig. 2C**, **Fig. S5C**). Meanwhile, EGFP⁻ embryonic microglia (yellow arrow) displayed robust expression of P2RY12 and TMEM119 protein throughout the study period (**Fig 2 B, C**).

FCRLS expression was assessed by flow cytometry in CX3CR1^{+/EGFP}::CCR2^{+/RFP} bone marrow chimeras. To distinguish cell populations, CX3CR1⁺ cells were labeled with an anti-CX3CR1 (PE-Cy7) antibody, defining embryonic microglia as PE-Cy7⁺GFP⁻ and infiltrating monocytes as PE-Cy7⁺GFP⁺ (**Fig. S6A**). The full representative gating strategy is detailed in **Figure S6B**. A Fluorescence Minus One (FMO) control was used to validate the specificity of the FCRLS staining. FCRLS, was absent on blood monocytes (grey) [7], but was robustly expressed by infiltrating monocytes (purple) as early as day 1 post-injury, at levels even higher than baseline microglia. This expression was sustained and eventually normalized to microglia levels by day 45 (**Fig. 2D, E**). Finally, the pan-myeloid marker IBA1 was expressed by 85% of infiltrating monocytes at day 1 (white arrow) and 100% thereafter (**Fig. S7**). Embryonic microglia (yellow arrow) also had robust expression of IBA1 throughout the study period (**Fig. S7**).

These findings were not specific to the injury model, as a chemically-induced retinal injury (NaIO₃) that damages the retinal pigment epithelium (RPE) [13] resulted in almost complete depletion of embryonic microglia, subsequent repopulation by peripheral monocytes that uniformly expressed P2RY12, TMEM119, and IBA1 (**Fig. S8**). This expression was sustained at 3 months, at which point cells have become amoeboid, with ~50% of them expressing MHC-II (**Fig. S8**).

This phenomenon was not restricted to the retina. In healthy, uninjured chimera mice, we observed that peripheral monocytes physiologically infiltrate and engraft into the brain and spinal cord over time. Within one month, these engrafted cells expressed TMEM119 and IBA1, and by one year, they robustly expressed P2RY12 as well, all while maintaining a quiescent, non-inflammatory (MHC-II negative) state and a ramified morphology indistinguishable from resident microglia (**Fig. S9, S10**).

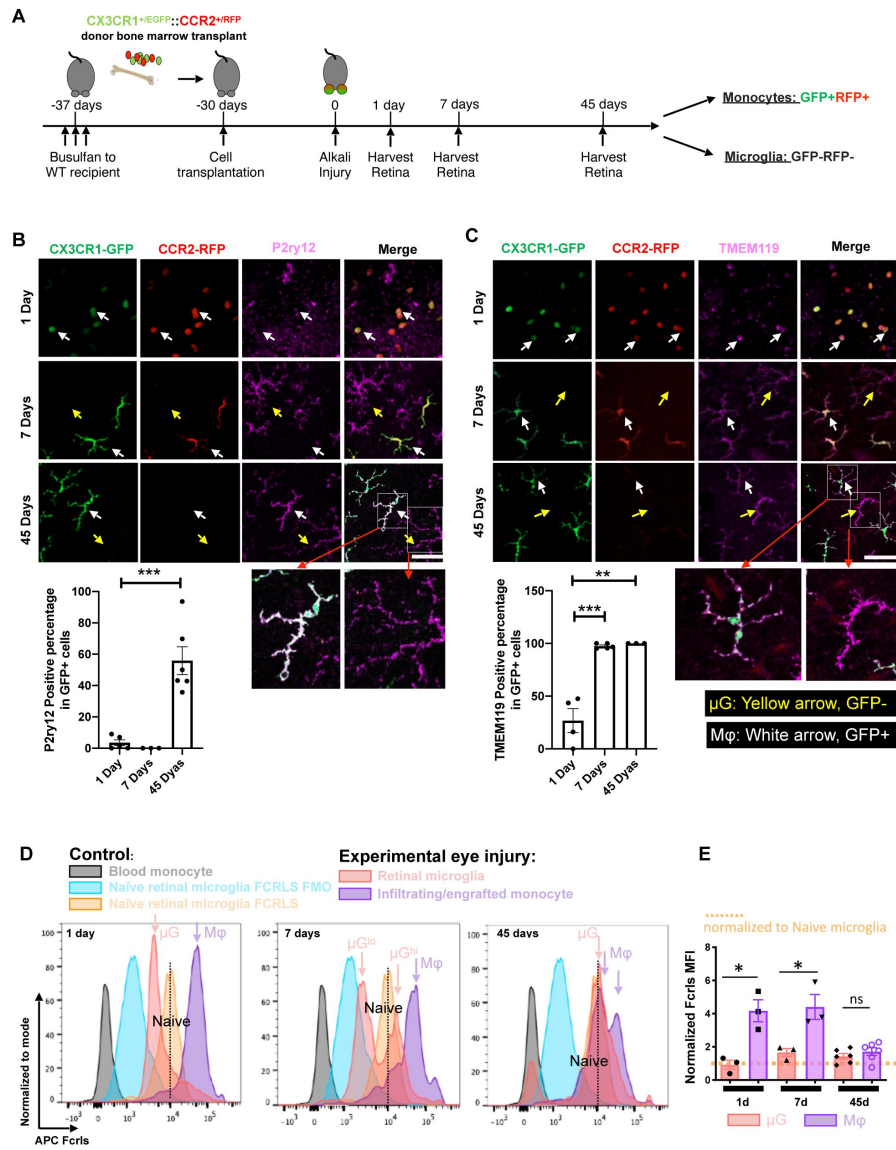


Figure 2.

Protein expression of P2RY12, TMEM119 and FCRLS by engrafted monocytes.

(A) CX3CR1⁺/EGFP⁺::CCR2⁺/RFP bone marrow chimeras were used to differentiate engrafted monocytes from embryonic microglia, followed by immunostaining and flow cytometry to assess expression of P2RY12, TMEM119, and FCRLS proteins. Engrafted monocytes were recognized as EGFP⁺ RFP⁺ cells while embryonic microglia were EGFP⁻ RFP⁻ cells. (B) P2RY12 is not expressed at day 1 and day 7 after monocyte infiltration into the retina, however 55% of EGFP⁺ engrafted monocytes showed positive expression of P2RY12 at day 45. ***, $P < 0.001$. (C) Twenty-five percent of EGFP⁺ peripheral monocytes (white arrow) expressed TMEM119 at day 1. By day 7, all GFP⁺ engrafted monocytes are TMEM119⁺. TMEM119 expression in engrafted monocytes is retained at day 45. (B, C) Data displayed as Mean $\pm$ SEM, independent t test ** $P < 0.01$; *** $P < 0.001$. Each dot represents 1 animal. Minimum of 3 animals per group. Yellow arrows indicate EGFP⁻ microglia, white arrows GFP⁺ engrafted monocytes. Scale bar = 50 μ m. (D-E) A CX3CR1⁺/EGFP⁺::CCR2⁺/RFP bone marrow chimera was employed to assess FCRLS expression in monocytes and microglia by flow cytometry. BMT CX3CR1⁺ cells were labeled with a conjugated antibody against CX3CR1 (APC) which allowed differentiation between embryonic microglia (APC⁺ EGFP⁻) and engrafted monocytes (APC⁺ EGFP⁺). Blood monocytes had no FCRLS expression (grey). One day after infiltration, APC⁺ EGFP⁺ peripheral monocytes (purple) acquired strong FCRLS expression, which was comparable to naïve embryonic microglia. FCRLS expression was retained by engrafted monocytes sustained at day 7. At day 45, FCRLS expression was similar to embryonic microglia in the same injured tissue or to naïve microglia. Mean $\pm$ SEM, independent t-test, * $P < 0.05$. Each dot represents 1 animal. Minimum of 3 animals per group. MFI: Median Fluorescence Intensity, Mφ: macrophages, μG: microglia.

Chromatin accessibility changes underpin the monocyte-to-microglia phenotypic switch

To investigate the epigenetic basis of this transformation [14–16], we performed ATAC-seq on flow-sorted circulating monocytes, naive microglia, and engrafted monocytes. We found that upon engraftment, monocytes undergo significant changes in chromatin accessibility that mirror those of resident microglia (Fig. 3). For the *p2ry12* and *tmem119* genes, specific chromatin peaks that were present in microglia but absent in circulating monocytes were gained by monocytes after they engrafted into the retina (Fig. 3B, D). For *fcr1l* and *aif1* (IBA1), chromatin was already accessible in circulating monocytes, suggesting these genes are epigenetically poised for rapid activation upon receiving the appropriate CNS signals (Fig. 3C, E).

A heatmap of all differential chromatin accessibility peaks clearly shows that engrafted monocytes at day 7 and 45 cluster epigenetically similar to microglia, rather than to their circulating precursors (Fig. 3F). Motif analysis of these differential peaks revealed a dramatic shift in the landscape of accessible transcription factor binding sites. Engrafted monocytes gained enrichment for motifs of hallmark microglia [17] transcription factors, including PU.1, CTCF, IRF, RUNX, and AP-1. Furthermore, they acquired motifs for disease-associated factors [18] like MITF and the pro-inflammatory factor NFκB1, which were not enriched in naive microglia, hinting at a distinct functional potential despite their morphological and marker-based resemblance (Fig. 4). Engrafted monocytes had enriched MAF and MEF motifs, absent in embryonic microglia. Microglia newly identified motifs in this study STAT1, FOXN1, KLFs, ATF3, and Npas4, were also enriched in engrafted monocytes (Fig. 4), suggesting a dynamic open chromatin accessibility shift in monocytes upon engraftment into the retina.

Engrafted Monocytes Downregulate Canonical Markers Ly6C and CD45

To evaluate the stability of canonical monocyte markers during chronic CNS pathology, we tracked the expression of Ly6C and CD45. It is well-established that Ly6C⁺ monocytes are recruited to the CNS during inflammation [7, 19], and that microglia can often be distinguished from monocytes based on their intermediate (CD45^{int}) versus high (CD45^{high}) expression of CD45 [20].

Using a CX3CR1^{+/EGFP}::CCR2^{+/RFP} bone marrow chimera model, we followed the phenotypic evolution of monocytes after engraftment into the retina (Fig. 5A). We observed a clear transition over 45 days: infiltrating monocytes, which were initially CCR2^{high}CX3CR1^{low} (Group 1), progressively matured into a predominantly CCR2^{low/-}CX3CR1^{high} population (Group 5), becoming phenotypically similar to resident microglia (Group 6) (Fig. 5B).

Ly6C expression was tightly coupled to this transition. Early infiltrating CCR2^{high} monocytes displayed high levels of Ly6C, which was then gradually downregulated as the majority of cells adopted the CCR2^{low/-}CX3CR1^{high} engrafted phenotype (Group 5) (Fig. 5C, D). Notably, a small subpopulation of engrafted monocytes that retained high CCR2 expression also maintained high Ly6C levels throughout the study period (Group 4) (Fig. 5C, D).

The expression of CD45 followed a similar pattern of repression. While the distinction between CD45^{high} monocytes and CD45^{int} microglia was clear at day 1 and 7, the majority of engrafted monocytes (88.5% of GFP⁺ cells) downregulated CD45 to intermediate levels by day 45, blurring the distinction between the two cell types (Fig. 5E–G).

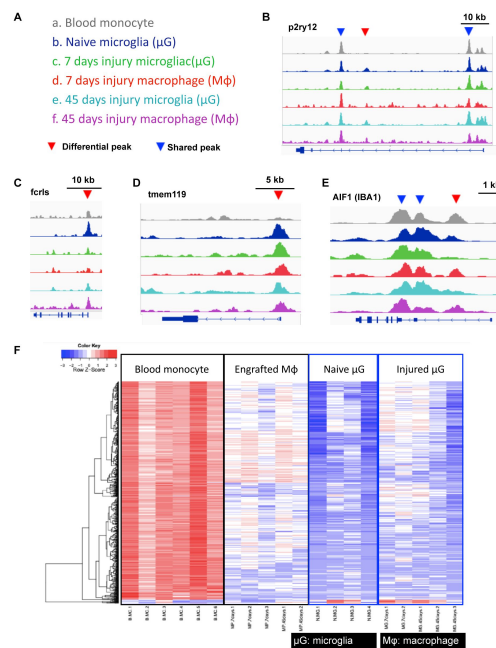


Figure 3.

ATAC-seq analysis to assess chromatin accessibility for *p2ry12*, *fcrls*, *aif1* (*iba1*), and *tmem119* genes in engrafted monocytes and microglia

In ATAC seq, peak width refers to the horizontal extent of a peak in the ATAC-seq signal track. It represents the range over which chromatin accessibility is elevated. A wider peak indicates a broader region of accessible chromatin. Peak amplitude refers to the height or intensity of the peak in the ATAC-seq signal track. It represents the level of chromatin accessibility at the peak's center. Higher amplitude suggests more frequent chromatin accessibility in that region. In the current study, circulating monocytes and naïve microglia were used as controls. Groups and color coding are listed in engrafted monocytes showed similar open chromatin peaks to circulating monocytes (blue arrows) but also displayed new open chromatin peaks not previously present in circulating monocytes (red arrows) but present in native (embryonic) microglia. One of the peaks under the *p2ry12* gene in naïve microglia (red arrow) was absent in circulating monocytes but acquired upon engraftment into the retina (**Fig. 3 B**). The *fcrls* gene had similar open chromatin peak across all groups (**Fig. 3 C**), though naïve microglia had wider and higher peak as compared to circulating monocytes (blue peak compared to grey). After engraftment, the amplitude of this peak increased in monocytes and became similar to microglia at day 45 (blue peak compared to purple), (**Fig. 3 C**). Similarly, *tmem119* chromatin was not accessible in circulating monocytes, however, it became accessible after monocyte engraftment into the retina (red arrow), (**Fig. 3 D**). Lastly, AIF1 gene (IBA1) had similar open chromatin peaks among the groups, corroborating the above transcriptional and protein findings showing gain of IBA1 expression by monocytes after engraftment into the retina (**Fig. 3 E**). (**A**) Color coding of analyzed groups. (**B**) *p2ry12* gene contains 3 open chromatin peaks, 2 of the peaks (blue arrowhead) are similar between the groups, but the 3rd peak is present in microglia (red arrowhead) but not in circulating monocytes. Upon engraftment into the retina, monocytes acquire the 3rd peak (red arrowhead) which is retained throughout the study period (45 days). (**C**) Open chromatin peaks for *fcrls* gene appear similar between the groups, with differences only in the amplitude of the peaks at 45 days in microglia and engrafted monocytes which have higher peaks compared to circulating monocytes or monocytes during early engraftment into the retina (7 days). (**D**) Open chromatin peaks for *aif1* gene (IBA1) appear similar between the groups, although microglia appeared to abolish one peak (red arrowhead) at 7 and 45 days after the injury. (**E**) *tmem119* has only one open chromatin peak, which is present in microglia but not in circulating monocytes, but upon engraftment, peripheral monocytes acquire this distinct peak (red arrowhead). (**F**) Heat map analysis of consensus peaks. Chromatin accessibility is scaled by row (Z-score) to emphasize relative differences across samples. Each row represents a genomic peak, and each column represents a biological replicate. Blood monocyte, n=6; Engrafted Mφ, n=5, including 3 samples of 7 days post injury and 2 samples of 45 days post injury; Naive µG, n=4; Injured µG, n=5, including 2 samples of 7 days post injury and 3 samples of 45 days post injury. The analysis shows engrafted monocytes at day 7 and 45 cluster more closely with microglia than with circulating blood monocytes, visually confirming a shift in the epigenetic landscape. Monocytes undergo significant open chromatin alterations upon engraftment into the retina which enables differentiation from circulating monocytes to tissue-phagocytes. Monocytes increase chromatin accessible for genes *p2ry12*, *tmem119*, *fcrls*, and *aif1* upon engraftment into the retina eventually acquiring a similar open chromatin signature to microglia.

(comparing μ G/ 7days M ϕ /45 days M ϕ to blood monocyte)					
Motif	Best Match	Naive μ G	7 day M ϕ	45 day M ϕ	
	PU.1	+	+	+	 Corroborate previous identified microglia motif
	CTCF	+	-	+	
	RUNX	+	+	-	
	AP-1	+	+	+	
	CEBP	+	-	-	
	IRF2	+	-	-	
	MAF	-	+	+	
	MEF	-	+	-	
	STAT1	+	+	+	 Newly identified motif
	FOXP1	+	-	+	
	KLF5/9	+	+	-	
	Atf3	+	-	-	
	Npas4	+	+	-	
	MITF	-	+	+	 Disease associated motif
	NFKB1	-	+	-	
Significant Motifs (E-values ≤ 0.05)					μ G: microglia M ϕ : macrophage

Figure 4.

ATAC-seq motif analysis for discovery of putative transcription factors regulating monocyte engraftment.

Motif analysis of differential open chromatin peaks identified by comparing naive microglia, retinal engrafted monocyte (7 and 45 days) compared to circulating monocytes. Blue section contains motifs assigned to transcription factors previously identified in human and mouse microglia, such as PU.1 (most common), CTCF, IRF, RUNX, MEF2, C/EBP, AP-1, MAF, and MEF. Green section contains enriched motifs assigned to novel transcription factors, such as STAT1, FOXP1, KLFs, ATF3, and Npas4. Red section contains previously reported disease associated motifs, such as MITF and NFKB1. Analysis of naive microglia identifies multiple reported factors but not MAF and MEF, which are identified only in engrafted monocytes. Engrafted monocyte at 7 and 45 days contain highly enriched motifs assigned to the above-mentioned transcription factors CEBP, IRF2, and ATF3. Disease-associated motifs assigned to MITF and NFKB1 are identified in engrafted monocytes but not in microglia. *E-value <0.05 for statistically significant motifs.*

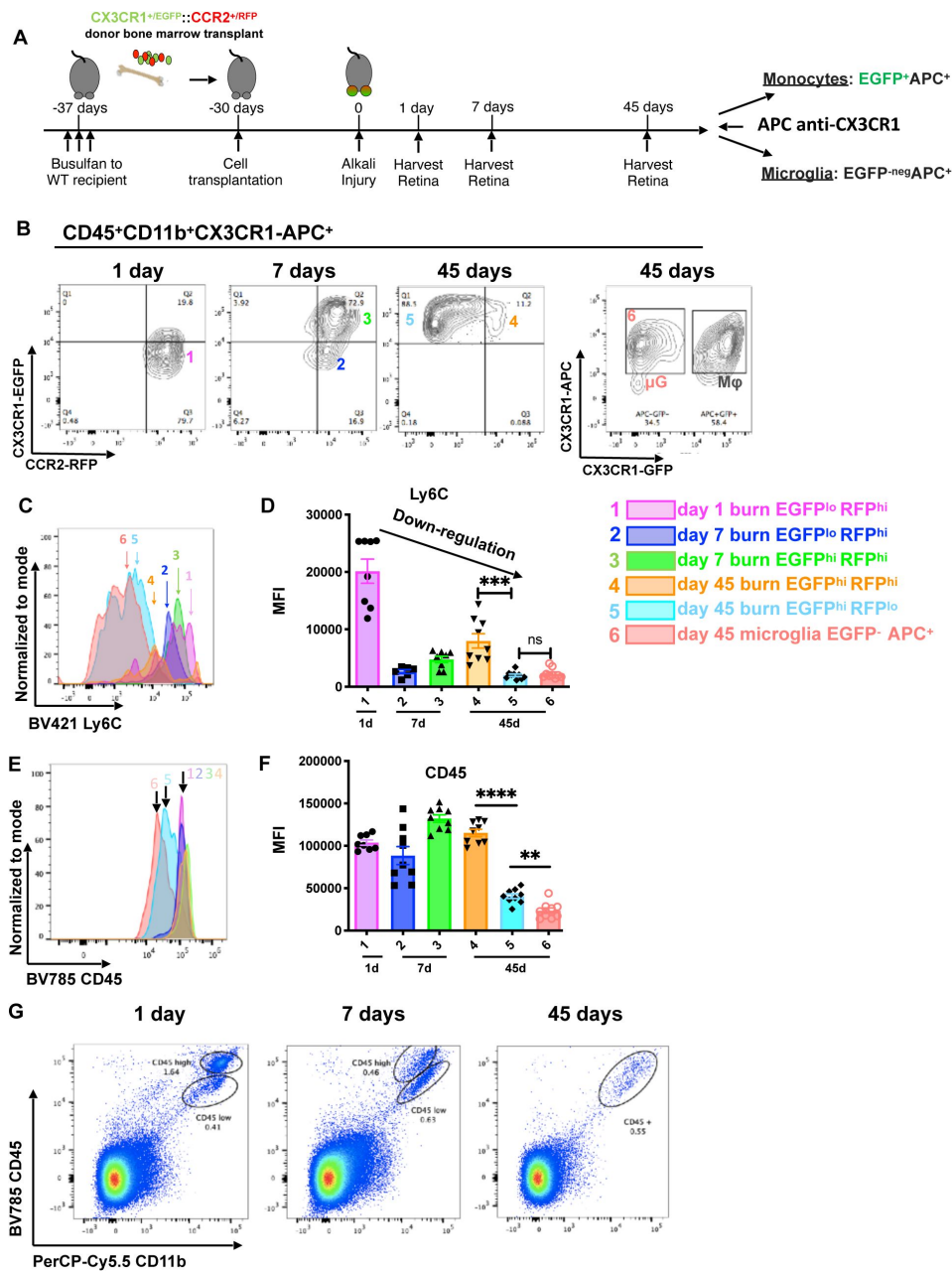


Figure 5.

Expression of conventional markers by microglia and engrafted monocytes.

(A) Development of a CX3CR1⁺/EGFP⁺:CCR2⁺/RFP bone marrow chimera model to differentiate microglia from peripheral monocytes following flow cytometry. Microglia: EGFP⁺ CX3CR1-APC⁺, engrafted monocytes: EGFP⁺CX3CR1-APC⁺. (B) Peripheral monocyte/macrophages repress CCR2 expression and enhance CX3CR1 expression during engraftment into the retina. Five distinct maturation phases of monocytes after engraftment are identified (Groups 1-5). A separate group of CCR2⁺CX3CR1⁺ cells, representing embryonic microglia (Group 6; μ G), is retained throughout the study period (45 days). (C-D) Engrafted monocytes have increase expression of Ly6C at day 1 of infiltration, which is gradually suppressed during engraftment. At 45 days, the majority of engrafted monocytes (Group 5) have similar Ly6C expression as microglia (Group 6). *ns*, not significant, independent t test, $***P < 0.001$. Each dot represents 1 animal. Minimum of 8 animals per group. (E-F) Engrafted monocytes exhibit sustained expression of CD45 at days 1 and 7 followed by repression in subpopulations of these cells (Group 5), and at day 45 reaching equal levels compared to retinal microglia (Group 6). Independent t test, $**P < 0.01$, $***P < 0.001$. Each dot represents 1 animal. Minimum of 8 animals per group. (G) A clear population of CD45^{high} and CD45^{int} is present at day 1 and 7 after injury, but not at day 45 due to CD45 repression in engrafted monocytes.

Collectively, these data demonstrate that in a chronic setting, the expression of both Ly6C and CD45 on engrafted monocytes converges with that of resident microglia. This renders these conventional markers unreliable for accurately discriminating between the two populations long-term. A comprehensive summary of these dynamic changes in morphology and marker expression over time is provided in **Figure 6**.

Discussion

The precise contribution of microglia to CNS pathology remains a subject of intense debate, with studies reporting both neuroprotective and deleterious roles [21–25]. This controversy is fueled by technical limitations and, as our study demonstrates, a critical lack of specific markers to differentiate resident microglia from peripherally-derived monocytes that infiltrate and permanently engraft into the CNS [2]. Here, we provide definitive evidence that engrafted monocytes are epigenetically reprogrammed to express a suite of markers—P2RY12, TMEM119, FCRL5, and Iba1—that are widely considered to be microglia-specific. This finding challenges the foundations of how these cell populations are identified and calls for a re-evaluation of their respective roles in CNS disease, in light of the more pro-inflammatory phenotype of monocytes [2, 26].

Our data directly refute the presumed specificity of TMEM119 and P2RY12. While TMEM119 was proposed as a stable marker to differentiate microglia from infiltrating monocytes [6], we show that CCR2⁺ monocytes begin expressing TMEM119 within a day of retinal infiltration, with ubiquitous expression in engrafted cells thereafter. Similarly, P2RY12 and FCRL5, which were initially characterized as absent from infiltrating monocytes in acute EAE models [7], are clearly upregulated in our chronic engraftment models across the retina, brain, and spinal cord. The expression of P2RY12 was notably heterogeneous and appeared later, suggesting a more complex and gradual maturation process. This dynamic and region-specific regulation was particularly evident in the spinal cord, where P2RY12 expression was absent in engrafted cells at one month but became robust by one year.

These findings have profound implications. For instance, in Alzheimer's disease research, P2RY12-negative myeloid cells surrounding Aβ plaques are often interpreted as dysfunctional microglia contributing to pathology [27, 28]. Our results raise the provocative possibility that this P2RY12-negative population could, in fact, be composed of recently engrafted monocytes. If true, this would fundamentally alter the interpretation of their role, shifting the focus from microglial failure to the pathological contribution of a distinct, peripherally-derived cell type. Likewise, the ubiquitous use of IBA1 [28–35] and CD45 [36] expression profiles to identify microglia in studies of tauopathy and multiple sclerosis must be reconsidered, as we show that long-term engrafted monocytes adopt an IBA1⁺ and CD45^{int} signature identical to that of resident microglia. Lastly, the use of expression of CD45^{hi} CD11b⁺ conventional marker is also inadequate, as engrafted monocytes have similar expression levels (CD45^{lo} CD11b⁺) at 45 days as microglia.

Mechanistically, our work reveals that this phenotypic convergence is driven by deep epigenetic remodeling. The ATAC-seq analysis provides a compelling explanation for the observed changes, showing that engrafted monocytes gain chromatin accessibility at key microglia identity gene loci (*P2ry12*, *Tmem119*). The motif analysis further solidifies this conclusion, revealing an enrichment for transcription factors like PU.1, IRF, and RUNX, which are known master regulators of microglial identity and microgliogenesis in humans and mice [17, 37]. Other established microglia genes, such as *SPP1*, *C1qa*, and *Ms4a7* [38] also become accessible after monocyte engraftment, a feature not observed in circulating blood monocytes. Critically, this analysis also provides a mechanistic clue to the functional divergence of these cells. The unique enrichment of motifs for the disease-associated transcription factor MITF and the pro-inflammatory hub NFκB1 in engrafted monocytes—but not in naive microglia—offers a powerful molecular explanation for

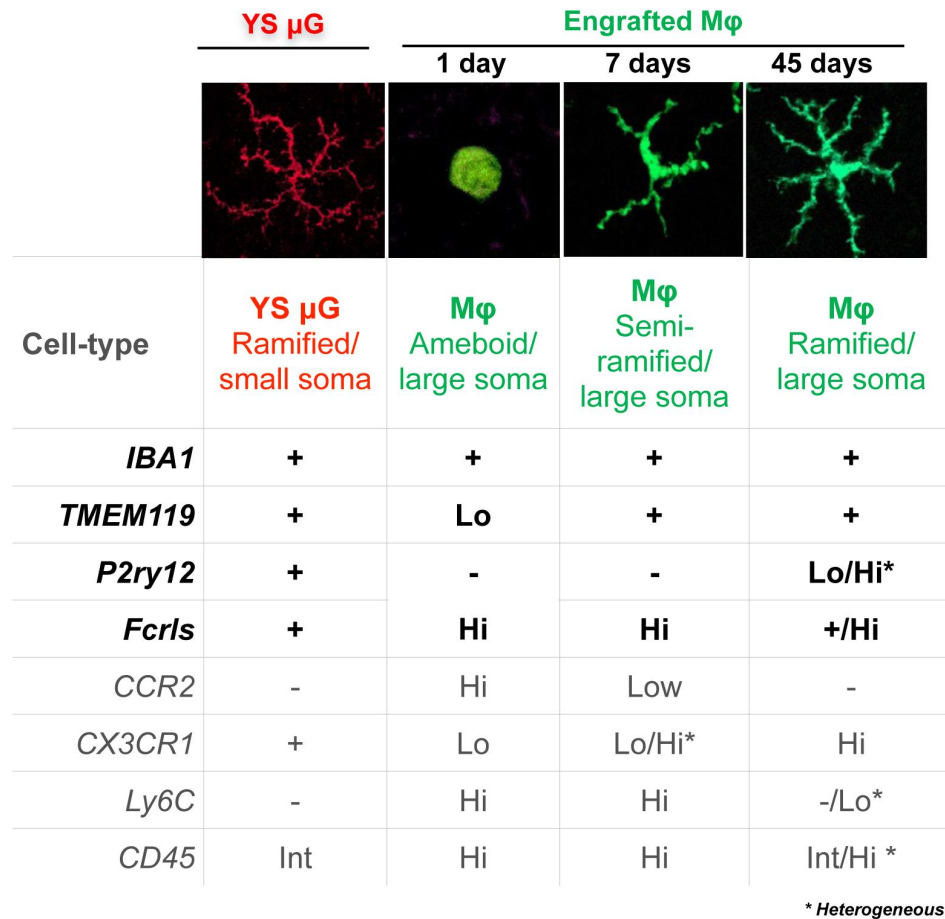


Figure 6.

Summary of the microglia and monocyte signature

Monocyte transition from highly amoeboid to highly ramified cells during engraftment into the retina, becoming morphometrically identical and indistinguishable from retinal microglia. These changes are accompanied by suppression of monocyte markers CCR2, Ly6C, and CD45, and upregulation of the tissue-resident macrophage marker CX3CR1^{+/GFP} and microglia markers IBA1, TMEM119, P2RY12, and FCRLS. P2RY12 appears to be conditionally specific to microglia during early infiltration of monocytes) and to a subpopulation of engrafted monocytes.

their previously reported pro-neurodegenerative phenotype [2, 18]. Thus, while they acquire the *look* of microglia, they appear epigenetically hardwired for a more aggressive, inflammatory response.

We also uncover a fundamental difference in immune surveillance across the CNS. In healthy animals, monocyte engraftment is a normal physiological process in the brain and spinal cord as shown in the current study, occurring steadily over the animal's lifespan, as well as the optic nerve as shown by our laboratory previously [2, 4]. In contrast, the healthy retina remains exclusively populated by embryonic microglia, requiring injury to trigger monocyte infiltration [2]. This distinction has major implications for understanding regional vulnerability and designing region-specific therapies. The state of cellular activation may be more critical than ontology; quiescent monocytes replacing microglia in the healthy brain are MHC-II negative, whereas injury-elicited monocytes in the retina are MHC-II positive and reactive (Fig. S8). This suggests that targeting the *function* of reactive monocytes, perhaps via epigenetic modulation, could be a powerful therapeutic strategy. This study has limitations. The bone marrow chimera model relies on myeloablation, which could potentially alter hematopoiesis, though we achieved near-complete chimerism with minimal barrier disruption [2]. Parabiosis models could offer an alternative, though more technically challenging to establish [39]. Cre-based fluorescent lineage tracing model could also be an alternative [8, 40], but incomplete activation or leaky Cre expression can lead to mislabeling of microglia and monocytes. Our model achieved near complete chimerism (~95%) and long-term engraftment of the donor cells into the CNS, which is essential for molecular profiling. While we provide robust ATAC-seq data, future studies using CUT&RUN [41] or ChIP-seq [17] could further dissect the specific histone modifications and enhancer landscapes driving this transformation. Future studies will need to investigate the functional dichotomy of these long term CNS engrafted monocytes and their ability to present as homeostatic microglia while primed for hyperinflammation. This potential hybrid state could explain the clinical difficulty of chronic neurodegenerative diseases.

In conclusion, our study demonstrates that the CNS microenvironment actively reshapes the identity of infiltrating monocytes, compelling them to adopt a microglia-like phenotype and express canonical microglia-specific markers. This mimicry invalidates their use for definitive lineage tracing in chronic disease settings. Until truly stable and specific protein markers are developed, the field must rely on sophisticated fate-mapping strategies and remain vigilant about the potential for cellular misidentification. Unraveling the functional consequences of this remarkable cellular plasticity will be key to understanding and ultimately treating a wide range of debilitating CNS diseases.

Materials and Methods

Animal Models and Procedures

All animal experiments were conducted in strict accordance with the Association for Research in Vision and Ophthalmology (ARVO) Statement for the Use of Animals in Ophthalmic and Vision Research and were approved by the Animal Care Committee of the Massachusetts Eye and Ear. Mice were bred and housed at the Massachusetts Eye and Ear Animal Facility and were used at 6–12 weeks of age.

Bone-Marrow Chimera Generation

To distinguish resident microglia from infiltrating peripheral monocytes, we generated bone marrow transplant (BMT) chimeras. Briefly, recipient C57BL/6J mice were myeloablated via intraperitoneal (i.p.) injections of busulfan (35 mg/kg; Sigma-Aldrich) administered 7, 5, and 3 days prior to transplantation. Bone marrow cells (5×10^6) were harvested from donor mice (either CX3CR1^{+/EGFP}::CCR2^{+/RFP} or CX3CR1^{+/EGFP} or CX3CR1^{CreER-EYFP}::ROSA26^{tdTomato}) and transferred

into the myelodepleted recipients via tail vein injection one month before experimental injury. To prevent infection, mice received Bactrim (trimethoprim/sulfamethoxazole) in their drinking water for 15 days following busulfan treatment. Recipient C57BL/6J (Stock No. 000664), Ccr2-RFP (Stock No. 017586), CX3CR1-EGFP (Stock No. 005582), Cx3cr1-CreER (Stock No. 021160), and ROSA26-tdTomato (Stock No. 007914) mice were obtained from The Jackson Laboratory. CX3CR1⁺/EGFP::CCR2⁺/RFP donor mice were generated by crossing Ccr2-RFP and CX3CR1-EGFP lines

Tamoxifen administration

For lineage tracing in the CX3CR1^{CreER-EYFP}::ROSA26^{tdTomato} chimera model, tamoxifen (20 mg/ml in corn oil) was administered two weeks after corneal injury, a time point allowing for monocyte infiltration and engraftment. Mice received two i.p. injections of tamoxifen at a dosage of 75 mg/kg, separated by one day, as previously described [42].

Mouse Model of Alkali Burn

One month after BMT, corneal alkali burns were induced as previously described. Briefly, mice were anesthetized with ketamine (60 mg/kg) and xylazine (6 mg/kg). Following application of topical proparacaine, a 2-mm filter paper soaked in 1 M NaOH was applied to the cornea for 20 seconds. The eye was immediately irrigated with sterile saline for 15 minutes. For post-procedural analgesia, Ethiqx XR (buprenorphine, 3.25 mg/kg, Covetrus North America, Cat: FP-001) was administered subcutaneously, and topical Polytrim antibiotic (polymyxin B/trimethoprim; Bausch & Lomb, Inc.) was applied to the eye.

NaIO₃ injury model

Retinal injury was also induced by a single i.p. injection of NaIO₃ (30 mg/kg; Sigma-Aldrich, Cat. S4007), per an established protocol. Animals were euthanized 14 days post-injection for analysis.

Flow Cytometry and Cell Sorting

Flow cytometry was used to analyze FCRLS, Ly6C, and CD45 expression and to sort cells for scRNA-seq and ATAC-seq. Retinas were collected from CX3CR1^{+/GFP}::CCR2^{+/RFP} chimeras at 1 day, 7 days, and 1.5 months post-injury and digested into single-cell suspensions using papain (Worthington Biochemical Corporation, Cat: LK003150). Infiltrating monocytes were identified as CD45⁺CD11b⁺GFP⁺ cells, while resident microglia were identified as CD45⁺CD11b⁺CX3CR1⁺GFP⁺ cells. After blocking with anti-CD16/32 (clone: 2.4G2), cells were stained with primary antibodies (Table 1) and analyzed on a BD FACS Aria[™] III cell sorter using FlowJo[™] 10 software.

Single cell RNAseq and gene expression profiling

Flow-sorted CD45⁺CD11b⁺ cells were encapsulated in microdroplets for single-cell library preparation. Using a high-throughput microfluidic platform, barcoded hydrogel beads containing reverse transcription reagents were picoinjected into the droplets to initiate in-drop cDNA synthesis. Libraries were prepared according to Klein et al. and sequenced on an Illumina HiSeq 2500 (100 bp paired-end) with approximately 1,000 cells per sample. Reads were mapped and quantified, and gene counts were normalized to reads per million (RPM).

RNA isolation and Quantitative real-time PCR analysis

A bone marrow transfer model was used to distinguish periphery infiltrated monocyte and resident microglia. Naïve microglia cells and blood monocyte were collected from uninjured bone marrow transferred mice. Injured microglia and engrafted monocyte were collected from retinas 45 days after ocular injury in bone marrow transferred mice. Cells were directly collected into 1ml Trizol (Thermo Fisher, Cat: 15596026). RNA extraction was performed per standard assay

ANTIBODY	FLUOROPHORE/SECONDARY	BRAND	CLONE	CATALOG NUMBER	DILUTION	APPLICATION
CD45	Brilliant Violet 785	Biolegend	30-F11	103149	1 to 100	Flow cytometry
CD11B	PerCP-Cyanine 5.5	Biolegend	M1/70	101228	1 to 100	Flow cytometry
LY6C	Brilliant Violet 421	Biolegend	HK1.4	128031	1 to 100	Flow cytometry
CX3CR1	APC	Biolegend	SA011F11	149008	1 to 100	Flow cytometry
CX3CR1	Brilliant Violet 605	Biolegend	SA011F11	149027	1 to 100	Flow cytometry
FCRL5	Alexa Fluor 647 donkey anti rat	Dr. Margeta			1 to 200	Flow cytometry
IBA1	Alexa Fluor 647 donkey anti rabbit	Wako		019-19741	1 to 300	Immunostaining
TMEM119	Alexa Fluor 647 donkey anti guinea pig	Synaptic systems		400 004	1 to 300	Immunostaining
P2RY12	Alexa Fluor 647 donkey anti rabbit	Dr. Margeta			1 to 200	Immunostaining
P2RY12	Alexa Fluor 647 donkey anti rabbit	Anaspec		as-55043a	1 to 200	Immunostaining
MHC-II	Alexa Fluor 647 donkey anti rat	BD Bioscience	I-A/I-E	556999	1 to 100	Immunostaining

Table 1.

Antibody information for flow Cytometry and immunostaining.

Both P2RY12 antibodies worked successfully in our experimental investigations.

recommendations Specifically, GlycoBlue™ Coprecipitant (Thermo Fisher, Cat: AM9515) was added in the RNA precipitation step to help visualize the RNA pellet. SMART-Seq V4 Ultra Low Input RNA Kit (Takara, Cat: 634890) was employed for RNA reverse transcription and cDNA amplification. RNA from around 400 cells was loaded in the experiment and 30ng cDNA could be yield after 18 cycles of cDNA amplification. cDNA amount was measured with Qubit dsDNA Quantification Assay kit (High sensitivity) (Thermo Fisher, Cat: Q32851). Quantitative real-time PCR analysis was conducted using TaqMan Probes and TaqMan universal PCR Master Mix (Thermo Fisher, Cat: 4304437). 250pg-500 pg cDNA was loaded for each PCR reaction.

ATAC-seq

The CX3CR1^{+/EGFP::CCR2^{+/RFP}} bone marrow transfer model was used to distinguish peripheral infiltrated monocyte and resident microglia. For flow sorting, infiltrated monocytes were gated as CD45⁺CD11b⁺GFP⁺ cells and microglia as CD45⁺CD11b⁺CX3CR1⁺GFP⁻. Each sample was pooled from 5 retinas (from 5 mice) and 1000 to 5000 cells collected and used for ATAC-seq analysis with the ATAC-seq kit from Active motif (Cat: 13150). Briefly, nuclei were isolated by adding 100 µL ice cold ATAC-lysis buffer and then incubated with the tagmentation master mix in a shaking heat block at 37°C/800 rpm for 30 min. Obtained DNA was purified and library generated by PCR reaction for 13 cycles using indexed primers according to the manufacturer's instructions. A quality control (QC) was performed to verify the size distribution of the PCR enriched library fragments. ATAC-seq sequencing was performed on an Illumina HiSeq 2000 instrument, resulting in 30 million paired-end 50 bp reads per sample. Reads were mapped to the mm9 reference mouse genome using BWA [43]. Those fragments with both ends unambiguously mapped to the genome that were longer than 100 bp were used for further analysis. Hotspot2 was used to detect significant peaks with FDR cutoff of 0.05 [44]. For the analysis of overlap between peak regions, we used a cutoff of 50% reciprocal overlap between the two compared regions. For the analysis of differential chromatin accessibility between groups of replicate samples, DiffBind R package was used: (<https://bioconductor.riken.jp/packages/3.2/bioc/vignettes/DiffBind/inst/doc/DiffBind.pdf>). Motif analysis was performed through MEME-CHIP (motif analysis of large nucleotide datasets).

Immunohistochemistry and Imaging

Mouse retinas were collected and prepared for flat mount staining 1, 7 and 45 days and 16 months after corneal alkali injury. Eyes were fixed in 4% paraformaldehyde overnight at 4°C then dissected. Mouse brain and spinal cord tissues were fixed in 4% paraformaldehyde overnight at 4°C and sections at 12 µm thickness. Immunostaining was performed by first blocking with PBS containing 5% normal donkey serum, 0.25% Triton-X-100 for 1 hour at room temperature, following by incubation of the tissues with primary antibody overnight or over the weekend at 4°C, and subsequent staining with secondary antibody, (Table 1). For retinal flat-mounts, whole retinas were laid flat after performing radial relaxing incisions and mounted on glass slides with coverslip. Images were captured using the Axio imager M2 (M2, Zeiss, Oberkochen, Germany).

Statistics

Results were analyzed with Prism 9 software. Medium fluorescence intensity (MFI) was used for quantification in flow cytometry experiments. Normality of data was assessed using the Shapiro-Wilk test (less than 50 samples) and independent Student's t test (unpaired, nonparametric or parametric depends on data type) was used to assess statistical significances between groups. Significance was assumed for P < 0.05.

Data availability statement

The data supporting the findings of this article are available within the article and/or its supplementary materials.

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Additional information

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J.L. and E.I.P. designed research; J.L., F.L., N.C. and B.Y. performed research; J.L., F.L., N.C., S.N. and J.S., V.C., S.N., D.V., D.W., R.M. and M.A.M. analyzed data; and J.L., B.Y., J.C., D.E.M., D.V., D.W., R.M. and E.I.P. wrote the paper.

Additional files

Supplementary figures [↗](#)

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Reviewer #1 (Public review):

Microglia are mononuclear phagocytes in the CNS and play essential roles in physiology and pathology. In some conditions, circulating monocytes may infiltrate in the CNS and differentiated into microglia or microglia-like cells. However, the specific mechanism is large unknown. In this study, the authors explored the epigenetic regulation of this process. The quality of this study will be significantly improved if a few questions are addressed.

(1) The capacity of circulating myeloid cell-derived microglia are controversial. In this study, the authors utilized CX3CR1-GFP/CCR2-DsRed (hetero) mice as a lineage tracing line. However, this animal line is not an appropriate approach for this purpose. For example, when the CX3CR1-GFP/CCR2-DsRed as the undifferentiated donor cell, they are GFP+ and DsRed+. When the cell fate has been changed to microglia, they will change into GFP+ and DsRed- cells. However, this process is mediated with busulfan and artificially introduced bone marrow cells in the circulating cell, which is not existed in physiological and pathological conditions. These artifacts will potentially bring in artifacts and confound the conclusion, as the classical wrong text book knowledge of the bone marrow derived microglia theory and subsequently corrected by Fabio Rossi lab^{1,2}. This is the most risk for drawing this conclusion. The top evidence is from the parabiosis animal model. Therefore, A parabiosis study before making this conclusion, combining a CX3CR1-GFP (hetero) mouse with a WT mouse without busulfan conditioning and looking at whether there are GFP+ microglia in the GFP- WT mouse brain. If there are no GFP+ microglia, the author should clarify this is not a physiological or pathological condition, but a defined artificial host condition, as previously study did³.

(2) In some conditions, peripheral myeloid cells can infiltrate and replace the brain microglia^{4,5}. Discuss it would be helpful to better understand the mechanism of microglia replacement.

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Reviewer #2 (Public review):

Mouse fate mapping studies have established that the bulk of microglia derives from cells that seed the brain early during development. However, monocytes were also shown to give rise to parenchymal CNS macrophages and thus are potential candidates for microglia replacement therapy. Whether monocyte-derived cells adopt bona fide microglia identities has remained under debate. The study of Liu et al addresses this important outstanding question, focusing on the retina.

Specifically, the authors investigate monocyte-derived macrophages that arise upon challenges in the murine retina using scRNAseq and ATACseq analyses, combined with flow cytometry and histology. They complement this approach with an analysis of BM chimeras and analyses of the latter. The authors conclude that monocyte-derived cells acquire markers that have originally been proposed to be microglia-specific, including P2ry12, Tmem119, and Fcrls.

In 2018, four comprehensive independent studies reported the analyses of monocyte-derived CNS macrophages (PMID 30451869, 30523248, 29643186, 29861285). Following transcriptome and epigenome analyses, these teams came to the collective conclusion that HSC-derived cells remain distinct from microglia. Using advanced fate mapping and better isolation and profiling tools, a more recent study, however, concluded that, if given sufficient time of CNS residence, most monocyte-derived macrophages can, at the transcriptome level, become essentially identical to microglia (PMID 40279248, <https://www.biorxiv.org/content/10.1101/2023.11.16.567402v1>).

Given this controversy, the study of Paschalis and colleagues, which focuses largely on retinal monocyte-derived cells, could have been a valuable resource and complement for clarification. Indeed, interestingly, their data suggest that microglia adaptation of monocyte-derived macrophages might be faster in the retina than in the CNS. However, for the reasons outlined below, the study falls in its present form short of providing significant new insight and is a missed opportunity.

Comments:

The major shortcoming of the study is that the authors decided to focus on a very limited number of genes to make their case, rather than performing a more informative, unbiased, and detailed global analysis. In contrast to what the authors state, much of the microglia community is, I believe, aware of experimental limitations and the problem with markers. Showing gain of microglia marker expression on monocyte-derived cells, or loss of monocyte markers, such as Ly6C, is not novel.

This is highlighted Fig. 3F. No one argues today that monocyte-derived tissue macrophages differ from blood monocytes (although the authors repeatedly emphasize this as novelty).

However, the heatmap shows that the engrafted cells clearly differ from naïve and injured microglia. What are these genes, their associated pathways ?

Also, how about expression of the *Sall1* gene that encodes a repressor that is considered important to maintain microglia identity (PMID37322178, 27776109). Somewhat surprisingly, *Sall1* was recently also shown to be expressed by monocyte-derived CNS macrophages (PMID 40279248). It would be valuable information if the authors can corroborate this finding.

The authors state in their discussion that monocyte-derived macrophages seem 'hardwired for inflammatory responses'. While this is an interesting suggestion, the NFkB motif enrichment is insufficient and should be complemented with a target list. Again, it would be important to be aware of heterogeneity.

A critical factor when analyzing CNS macrophages is the exclusion of perivascular CNS border-associated cells, which also holds for the retina (see PMID 38596358). This should be addressed. Can the authors discriminate BAM from microglia in their scRNAseq data set, for instance, by their CD206 expression or other markers ? BAM have been shown to display distinct transcriptomes and even as a contamination could introduce significant bias.

Even for the genes the authors focus on, it is hard to understand from the way the authors present the data what fraction of cells are positive. This would be critical information since there could be some heterogeneity. Flowcytometry analysis, including double staining for P2ry12, Tmem119, and Fcrls to see correlations, would here be valuable.

The authors state in their title that 'epigenetic adaptation drives monocyte differentiation'. However, since all gene expression is governed by the epigenome, this is trivial. I would argue that to gain meaningful insight and justify such a statement, it would require an in-depth global comparative analysis of the chromatin status of yolk sac microglia and monocyte-derived CNS macrophages, including CUT&RUN analysis for specific histone marks and methylation patterns.

Please cite and discuss PMID 30451869, 30523248, 29643186, 29861285, and in particular the more recent highly relevant study PMID 40279248.

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